

Propagation of sound and the assignment of shape factors in model porous materials having simple pore geometries

Michael R. Stinson and Yvan Champoux^{a)}

Institute for Microstructural Research, National Research Council, Ottawa, Ontario K1A 0R6, Canada

(Received 25 March 1991; accepted for publication 23 October 1991)

The acoustical properties of a class of simple porous materials have been studied experimentally and theoretically. Rigid-frame materials containing air-filled pores of uniform cross section were investigated. Two model porous materials were constructed, one with pores of rectangular cross section (0.0146×0.0172 cm) and one with pores of triangular cross section (0.037-cm sides). The characteristic impedance and propagation constant were measured for frequencies between 50 and 4500 Hz and good agreement with exact theoretical predictions was obtained. The exact theoretical expressions for specific pore shapes (e.g., slitlike, square, and triangular) can be used to investigate the assignment of shape factors by various general but approximate theoretical models. It is demonstrated that the model introduced by Attenborough requires a shape factor that is frequency dependent. A theoretical model, appropriate for materials containing pores of uniform cross section, that correctly treats the low- and high-frequency behavior, is presented. Given subsidiary measurements of flow resistivity, porosity, and tortuosity, a *single* shape factor leads to very good agreement with the exact solutions for all frequencies.

PACS numbers: 43.20.Gp, 43.20.Jr, 43.55.Ev

INTRODUCTION

The acoustical properties of rigid-frame, air-filled porous materials are of interest in many areas, particularly noise control applications. The development of materials with specialized properties requires a continuing refinement of theoretical ideas.

The empirical expressions presented by Delany and Bazley¹ permit estimates of the characteristic impedance and propagation constant of porous materials to be obtained if subsidiary measurements of the flow resistivity can be made. These estimates may be adequate for many applications. However, the expressions for characteristic impedance and propagation constant were deduced empirically and, indeed, they are not physical. The theoretical modeling of Morse and Ingard² is based on physically sound principles but does not deal directly with the microstructure of particular materials. As a result, the effective resistivity and stiffness of the air in the pores will have frequency dependences that must be determined experimentally.

In Attenborough's theoretical model,³⁻⁵ the pore microstructure is treated in considerably more detail. Viscous and thermal effects in individual pores are described separately⁶ through complex density and compressibility functions, using the exact solutions for uniform pores of circular cross section as a basis. The flow resistivity, porosity, and tortuosity are closely related to the size and distribution of pores and, if measured independently (through nonacoustical means), can be utilized as input to the theory. The influ-

ence of the cross-sectional shape of the pore (i.e., the deviation from circular) is treated by introducing appropriate *shape factors* that scale the arguments of the basis functions for complex density and compressibility. The "pore shape factor ratio" would, ideally, be a constant for a given type of material. As noted by Attenborough⁵ and as will be clear later in this article, however, this shape factor can vary significantly with frequency.

We can imagine porous materials having relatively simple microstructure, simple enough that analytic equations of acoustical propagation can be written down. Then, the exact predictions for these materials could be compared directly to the predictions that would be made by various general models (in particular, that of Attenborough). This comparison would provide a direct evaluation of the use of shape factors. Exact theoretical expressions exist for sound propagation in circular tubes and in tubes that are essentially "narrow slits" in cross section. Hypothetical porous materials based on these pore shapes are then convenient test objects, as indicated by their extensive use by Attenborough and Biot.^{7,8} Other pore geometries are possible, though. Recent work on sound propagation in uniform tubes⁹ provides a procedure for deriving the acoustical properties of pores having arbitrary cross-sectional shape. Exact expressions have been found for pores of rectangular cross section^{9,10} and pores of triangular cross section.¹¹ With these pore shapes, the use of shape factors can be examined in a more exacting manner than previously possible.

In this paper, the fabrication of two *model* porous materials, one containing rectangular pores and the other containing triangular pores, will be described. The measured acoustical properties of these samples will be used to validate

^{a)} On leave from the Mechanical Engineering Department, Faculty of Applied Science, Université de Sherbrooke, Sherbrooke, Québec J1K 2R1, Canada.

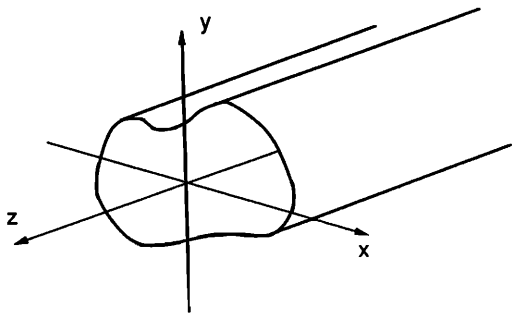


FIG. 1. A pore of arbitrary cross-sectional shape, uniform along its length, is considered. A rectangular coordinate system is introduced, with z along the pore direction.

the exact theoretical expressions appropriate for the two pore shapes. The exact expressions will, in turn, be contrasted with the predictions of Attenborough's theoretical model, and his use of shape factors examined. Finally, a new theoretical model that treats porous material containing pores of uniform (but arbitrary) cross section will be explored.

I. THEORY

A. Propagation in single tubes of uniform cross section

Consider a uniform tube, as sketched in Fig. 1, with the rectilinear coordinates x, y, z indicated. The cross-sectional shape is arbitrary, but constant along the length of the tube. The propagation of sound is controlled by both viscous and thermal effects within the tube walls. These effects may be treated separately,^{6,9,12} and we will consider both in turn.

The viscosity of the air within the tube enters the general force equation. From Stinson,⁹ the particle velocity in the z direction is given by

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} - \frac{i\rho_0\omega}{\eta} v = \frac{1}{\eta} \frac{dp}{dz}, \quad (1)$$

where ρ_0 and η are the density and viscosity of air and ω is the angular frequency. The pressure p depends only on the longitudinal position z . The solution v that is obtained, whether numerically or analytically, is a function of x and y coordinates, and equal to zero at the tube walls. We are primarily interested in the average velocity through the pore that is calculated using

$$\langle v \rangle = \frac{1}{A} \oint_A v(x, y) dx dy, \quad (2)$$

where A is the cross-sectional area of the tube. A complex density function $\rho(\omega)$ is introduced, relating the average velocity to the applied pressure gradient

$$i\omega\rho(\omega)\langle v \rangle = -\frac{dp}{dz}. \quad (3)$$

Because of the nonzero thermal conductivity of air, some heat is lost through the tube walls. The excess temperature τ (i.e., the difference between actual and ambient temperatures) is a solution of⁹

$$\frac{\partial^2 \tau}{\partial x^2} + \frac{\partial^2 \tau}{\partial y^2} - \frac{i\rho_0\omega C_p}{\kappa} \tau = -\frac{i\omega}{\kappa} p, \quad (4)$$

where C_p is the specific heat at constant pressure and κ is the thermal conductivity of air. The excess temperature is set equal to zero at the tube wall. Once a solution is determined, the average excess temperature $\langle \tau \rangle$ within the pore is calculated, in a manner equivalent to Eq. (2). This average is related to the local sound pressure through a complex compressibility function $C(\omega)$ given by

$$C(\omega) = (\langle \delta \rangle / \rho_0) / p = 1/P_0 - (\langle \tau \rangle / T_0) / p. \quad (5)$$

Here, P_0 and T_0 are the ambient air pressure and temperature, and the average excess density $\langle \delta \rangle$ has been related to $\langle \tau \rangle$ using the ideal gas law.

The equations from which the viscous and thermal effects are calculated, Eqs. (1) and (4), respectively, are identical in form. This equivalence in form leads to a general relationship between complex density and compressibility functions. Suppose that the solution for Eq. (1) is $v = \psi(x, y, \omega)$; the dependence on ω is explicitly noted. Then, it is easily verified by substitution that

$$\tau = -\frac{i\omega\eta}{\kappa} \frac{p}{(dp/dz)} \psi(x, y, N_{pr}\omega),$$

where $N_{pr} = C_p\eta/\kappa$ is the Prandtl number, is a solution of Eq. (4). Averaging over the cross section, we obtain

$$\langle \tau(\omega) \rangle = -\frac{i\omega\eta}{\kappa} \frac{p}{(dp/dz)} \langle \psi(N_{pr}\omega) \rangle. \quad (6)$$

For an ideal gas,

$$C_p - C_v = P_0 / (\rho_0 T_0), \quad (7)$$

where C_v is the specific heat at constant volume. Combining Eqs. (3), (5)–(7) and letting $\gamma = C_p/C_v$, we obtain

$$C(\omega) = (1/\gamma P_0) \{ \gamma - (\gamma - 1) [\rho_0 / \rho(N_{pr}\omega)] \}. \quad (8)$$

This relation is generally true, valid for all *uniform* tubes.

B. Solutions for specific geometries

For certain simple pore shapes, exact solutions may be written down. The shapes that will be considered are shown in Fig. 2 and their geometric descriptors noted.

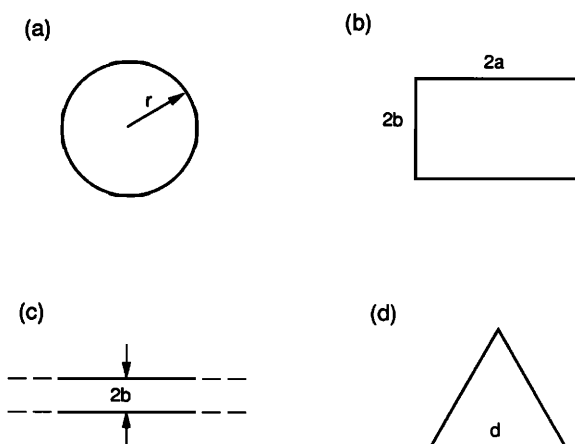


FIG. 2. The pore cross-sectional shapes for which exact, analytic expressions are obtained. They are (a) circle of radius r , (b) rectangle of half-width b and half-length a , (c) slit of half-width b , and (d) equilateral triangle with sides d .

First, for a circular tube of radius r the complex density function is well-known, and is given by⁶

$$\rho_c(\omega) = \rho_0 \left[1 - 2T(\sqrt{-i\lambda_c})/\sqrt{-i\lambda_c} \right]^{-1}, \quad (9)$$

where

$$T(\xi) = J_1(\xi)/J_0(\xi), \quad (10)$$

J_0 and J_1 are Bessel functions of zero and first order, and the parameter λ_c is given by

$$\lambda_c = r(\rho_0\omega/\eta)^{1/2}. \quad (11)$$

The complex compressibility function may be obtained by applying Eq. (8) leading to the result obtained originally by Zwikker and Kosten⁶

$$C_c(\omega) = (1/\gamma P_0) \{ 1 + (\gamma - 1) [2T(\sqrt{-iN_{pr}\lambda_c})/\sqrt{-iN_{pr}\lambda_c}] \}. \quad (12)$$

Next, we consider a pore of rectangular cross section, having half-width b and half-length a . Complex density and compressibility functions have been derived recently:^{9,10}

$$\rho_r(\omega) = \rho_0 \frac{\eta a^2 b^2}{4i\omega\rho_0} \times \left(\sum_{k=0}^{\infty} \sum_{n=0}^{\infty} \frac{1}{\alpha_k^2 \beta_n^2 (\alpha_k^2 + \beta_n^2 + i\omega\rho_0/\eta)} \right)^{-1}, \quad (13)$$

$$C_r(\omega) = \frac{1}{\gamma P_0} \left(\gamma - (\gamma - 1) \frac{4i\omega N_{pr}\rho_0}{\eta a^2 b^2} \times \sum_{k=0}^{\infty} \sum_{n=0}^{\infty} \frac{1}{\alpha_k^2 \beta_n^2 (\alpha_k^2 + \beta_n^2 + i\omega\rho_0 N_{pr}/\eta)} \right), \quad (14)$$

where the coefficients α_k and β_n are given by

$$\alpha_k = (k + 1/2)\pi/a, \quad (15)$$

$$\beta_n = (n + 1/2)\pi/b.$$

A limiting case of the rectangular pore is a slit, of half-width b , as indicated in Fig. 2(c). Letting $a \rightarrow \infty$, Eqs. (13) and (14) can be evaluated to obtain

$$\rho_s(\omega) = \rho_0 \left[1 - \tanh(\sqrt{i\lambda_s})/\sqrt{i\lambda_s} \right]^{-1}, \quad (16)$$

$$C_s(\omega) = (1/\gamma P_0) \{ 1 + (\gamma - 1) [\tanh(\sqrt{iN_{pr}\lambda_s})/\sqrt{iN_{pr}\lambda_s}] \}, \quad (17)$$

where the parameter λ_s is given by

$$\lambda_s = b(\omega\rho_0/\eta)^{1/2}. \quad (18)$$

Equations (16)–(18) were discussed by Attenborough.³

The final pore shape that will be considered is an equilateral triangle of side d , as indicated in Fig. 2(d). The derivation of the complex density function will be described in somewhat more detail since there seems to be no previous solution in the literature. We require a solution of Eq. (1) which has $v = 0$ on the three sides and has three-fold symmetry. An expansion of the form

$$v(x, y) = \sum_{n=1}^{\infty} B_n f_n(x, y), \quad (19)$$

where

$$f_n(x, y) = \sin\left(\frac{2\beta_n y}{\sqrt{3}}\right) \sin \beta_n \left(x + \frac{y}{\sqrt{3}}\right) \sin \beta_n \left(x - \frac{y}{\sqrt{3}}\right) \\ = \frac{1}{4} \left[\sin\left(\frac{4\beta_n y}{\sqrt{3}}\right) - \sin 2\beta_n \left(x + \frac{y}{\sqrt{3}}\right) \right. \\ \left. + \sin 2\beta_n \left(x - \frac{y}{\sqrt{3}}\right) \right] \quad (20)$$

is assumed, with

$$\beta_n = n\pi/d. \quad (21)$$

Substituting this expansion into Eq. (1) gives

$$\sum_{n=1}^{\infty} B_n \left(\frac{16\beta_n^2}{3} + \frac{i\omega\rho_0}{\eta} \right) f_n(x, y) = \frac{1}{\eta} \frac{dp}{dz}. \quad (22)$$

It may be shown that the functions $f_n(x, y)$ are orthogonal in that

$$\oint_A f_n(x, y) f_{n'}(x, y) dx dy = \frac{3\sqrt{3}}{128} d^2 \delta_{n, n'}, \quad (23)$$

where A is the cross section of the triangular tube. As well,

$$\oint_A f_n(x, y) dx dy = -\frac{3\sqrt{3}}{16n\pi} d^2. \quad (24)$$

These results may be used to determine the coefficients B_n . Both sides of Eq. (22) are multiplied by $f_{n'}(x, y)$ and integrated over the cross section, leading to

$$B_n = -\frac{dp}{dz} \frac{8}{\beta_n d (16\eta\beta_n^2/3 + i\omega\rho_0)}, \quad (25)$$

and, hence,

$$v(x, y) = -2 \frac{dp}{dz} \sum_{n=1}^{\infty} \frac{\sin 4\beta_n y/\sqrt{3} - \sin 2\beta_n (x + y/\sqrt{3}) + \sin 2\beta_n (x - y/\sqrt{3})}{\beta_n d (16\eta\beta_n^2/3 + i\omega\rho_0)}. \quad (26)$$

The sums shown in Eq. (26) can be evaluated by making use of the relations¹³

$$\sum_{n=1}^{\infty} \frac{\sin n\xi}{n} = \frac{\pi - \xi}{2}, \quad 0 < \xi < 2\pi, \quad (27)$$

$$\sum_{n=1}^{\infty} \frac{n \sin n\xi}{n^2 + \alpha^2} = \frac{\pi \sinh \alpha(\pi - \xi)}{2 \sinh \alpha\pi}, \quad 0 < \xi < 2\pi.$$

We obtain a velocity distribution given by

$$v(x, y) = \frac{dp}{dz} \left[\sinh \epsilon + \sinh \epsilon \left(1 - \frac{4y}{\sqrt{3}d} \right) \right. \\ \left. - \sinh \epsilon \left(1 - \frac{2(x + y/\sqrt{3})}{d} \right) \right. \\ \left. - \sinh \epsilon \left(1 + \frac{2(x - y/\sqrt{3})}{d} \right) \right] (i\omega\rho_0 \sinh \epsilon)^{-1}, \quad (28)$$

where the parameter ϵ is given by

$$\epsilon^2 = \frac{3}{16} \frac{i\omega\rho_0 d^2}{\eta}. \quad (29)$$

The expression for $v(x,y)$ may be integrated over the cross section, according to Eq. (2), to give the average velocity $\langle v \rangle$. Combining this result with the definition for the complex density function leads to

$$\rho_t(\omega) = \rho_0 \epsilon^2 / (\epsilon^2 - 3\epsilon \coth \epsilon + 3). \quad (30)$$

And, from the general relation Eq. (8), the complex compressibility function is

$$C_t(\omega) = (1/\gamma P_0) \{ \gamma - (\gamma - 1) [(N_{pr} \epsilon^2 - 3N_{pr}^{1/2} \epsilon \coth N_{pr}^{1/2} \epsilon + 3) / N_{pr} \epsilon^2] \}. \quad (31)$$

C. Extension to bulk materials

We will consider a rigid frame material containing pores of uniform cross section. The fraction Ω of the overall volume taken up by the pores is the porosity. Moreover, we suppose that the pores may not be straight. Let q be the ratio between pore length and sample length. For this simple system, the tortuosity is q^2 . However, it is noted that in samples having pores of varying cross-sectional area, the tortuosity would be more complicated than just this simple geometrical measure of the sinuosity of the pores. Let the complex density and compressibility functions corresponding to a single pore be given by $\rho_x(\omega)$ and $C_x(\omega)$, respectively.

The complex density of this bulk material is defined as

$$i\omega\rho_b(\omega)v_b = -\frac{dp}{dz}, \quad (32)$$

where v_b is the average fluid velocity, obtained by averaging the actual velocity over the *entire* cross section. This is consistent with the complex density function discussed by Attenborough.³⁻⁵ We obtain

$$\rho_b(\omega) = (q^2/\Omega)\rho_x(\omega). \quad (33)$$

The complex compressibility function for the bulk material can be defined by

$$C_b(\omega) = (\delta_b/\rho_0)/p, \quad (34)$$

where δ_b is the excess fluid density averaged over the entire cross section. It is related to the single pore compressibility function by

$$C_b(\omega) = \Omega C_x(\omega). \quad (35)$$

The characteristic impedance of the bulk material may be shown to be

$$Z = [\rho_b(\omega)/C_b(\omega)]^{1/2} \quad (36)$$

and the propagation constant,

$$m = i\omega[\rho_b(\omega)C_b(\omega)]^{1/2}. \quad (37)$$

II. EXPERIMENTAL PROCEDURE

A. Measurement systems

The acoustical measurement system has been described elsewhere¹⁴ and will be discussed only briefly here. Essen-

tially, the input impedance at the surface of a sample and the pressure transfer function along the length of the sample are measured. From these two quantities (knowing the lengths of the sample and the terminating cavity) both the characteristic impedance Z and propagation constant m of the bulk material may be calculated. (A similar approach has been described in the recent article by McIntosh *et al.*¹⁵) From Eqs. (36) and (37), the corresponding complex density and compressibility functions, for the bulk material, are

$$\rho_b(\omega) = \pm mZ/i\omega \quad (38)$$

and

$$C_b(\omega) = \pm m/i\omega Z. \quad (39)$$

The complex density and compressibility functions corresponding to a single pore can be calculated using Eqs. (33) and (35).

The surface impedance was determined in two ways. At the higher measurement frequencies, the standing wave pattern was determined using a movable probe tube.¹⁴ At the lower frequencies, the technique introduced recently by Gibiat¹⁶ was found, at least for the model porous materials considered here, to give somewhat smaller experimental uncertainties. The sound tubes used for these measurements have an inner diameter of 4.471 cm.

Measurements were also made of the dc flow resistivity¹⁷ and the sample porosity.¹⁸ These permitted a critical verification of the pore dimensions.

B. Model porous materials

Two model porous materials were built. One sample contains pores of rectangular cross section and the other, pores of triangular cross section. Both were fabricated by machining parallel slots of the appropriate shape in sheets of Plexiglas and stacking many sheets to build up the sample. Figure 3 shows, schematically, portions of a few sheets of each type, illustrating how the samples were constructed.

Each Plexiglas sheet was initially machined to a thickness of 0.020 in. A total of 100 sheets, each 10.2×14.52 cm, were prepared and clamped tightly together. To achieve an effectively circular sample continuous with the sound tube of the measurement system, pores were uniformly distributed over a circular region (as viewed from the end of the sample, along the pore direction) having a diameter of 4.471 cm. Hence, the number of pores machined in each sheet varied. To cut the rectangular pores, a slit saw of nominal 0.006-in. width was used, cutting to a target depth of 0.006 in. A total of 12 111 pores were machined in this sample. For the triangular pores, a bit with a 60-deg point was used, cutting to a target depth of 0.0105 in. The final sample contains a total of 6023 pores.

For the application of the theoretical expressions, an accurate estimate of the dimensions of the pores is required. However, the dimensions could not be determined with sufficient accuracy from direct observations. For example, for the rectangular pores, measured dimensions of $(0.015 \pm 0.003) \times (0.017 \pm 0.003)$ cm were obtained using a traveling microscope technique. With these uncertainties, the calculated dc flow resistivity could be anywhere between

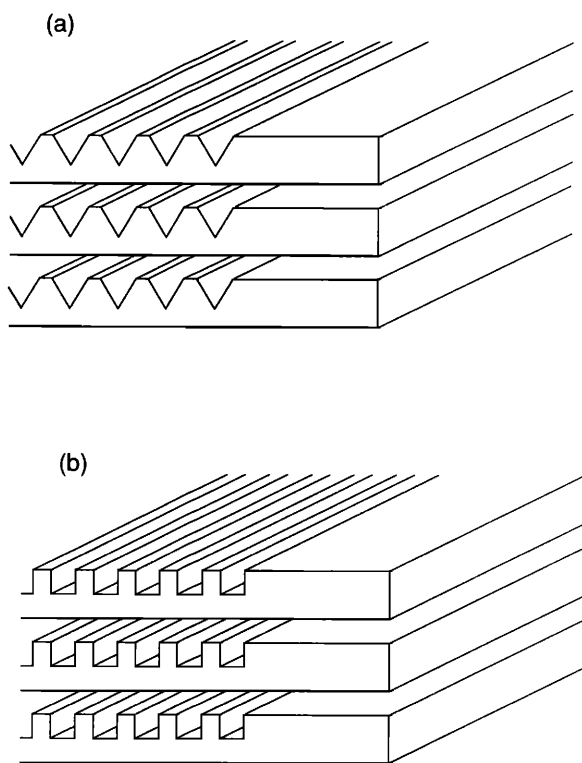


FIG. 3. Sketch of the procedure used to fabricate model porous materials. Straight, parallel grooves of (a) triangular or (b) rectangular cross section are machined in Plexiglas sheets. Many such sheets are stacked and held together to form the samples. One sample contained 6023 triangular pores, each 0.0374 cm on a side. The other sample contained 12111 rectangular pores, each 0.0146×0.0172 cm.

51 and 235 cgs rayl/cm. The flow resistivity, though, was measured independently, giving a value (107.6 ± 1.7) cgs rayl/cm. Hence, we can work backwards using this relatively precise determination to infer the pore dimensions. A pore cross section of 0.0146×0.0172 cm was adopted. This choice gives a sample porosity of 0.194 and a dc flow resistivity of 107.5 cgs rayl/cm.

From microscopic observation of the triangular pore sample, it is noted that the pores are approximately equilateral in cross section, with some rounding (0.001- to 0.002-in. radius) in one of the three vertices. The base of the triangle has an observed length of (0.034 ± 0.003) cm. The uncertainty in this measurement allows considerable range in calculated flow resistivity, as was the case for the rectangular pore sample. We will adopt a side dimension of 0.0374 cm. This choice gives a porosity of 0.232 and a flow resistivity of 44.7 ± 0.7 cgs rayl/cm, compatible with the measured flow resistivity of (45.0 ± 0.7) cgs rayl/cm.

The porosities of the samples were also measured. A value of 0.225 was found for the rectangular pore material, and a value of 0.25, for the triangular pore material. These values are larger than expected from the assumed dimensions. It is believed, though, that small spaces between the stacked Plexiglas sheets and tiny transverse grooves on the surfaces of the sheets (remnants from the machining process) can account for these differences.

III. EXPERIMENTAL RESULTS

The acoustical properties of the triangular pore model material are summarized in Fig. 4. The characteristic imped-

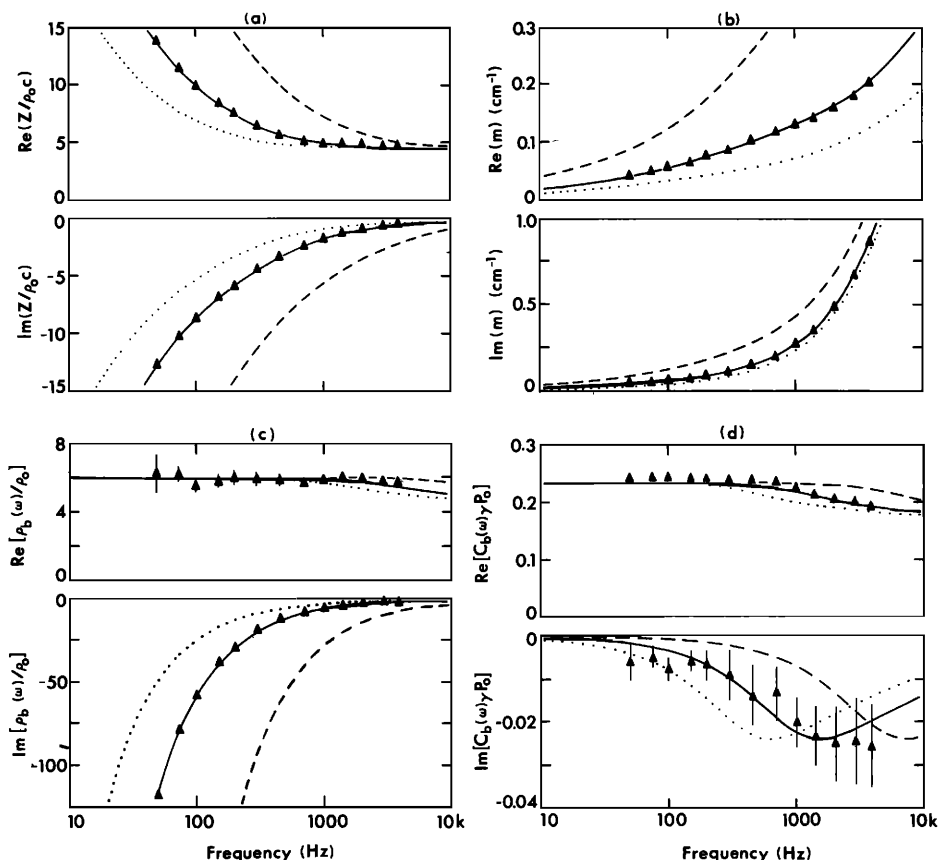


FIG. 4. Comparison of measured (triangular symbols) and theoretical (solid curves) acoustical properties for a porous sample containing 6023 uniform triangular pores, each with sides of 0.0374 cm, giving a porosity of 0.232. The four panels show the real and imaginary components of (a) the characteristic impedance Z , (b) the propagation constant m , (c) the complex density function $\rho_b(\omega)$, and (d) the complex compressibility function $C_b(\omega)$, of the bulk material. The dashed and dotted curves are the theoretical predictions that would be obtained if the triangular pores had sides of 0.017 and 0.057 cm, respectively, with the same porosity of 0.232.

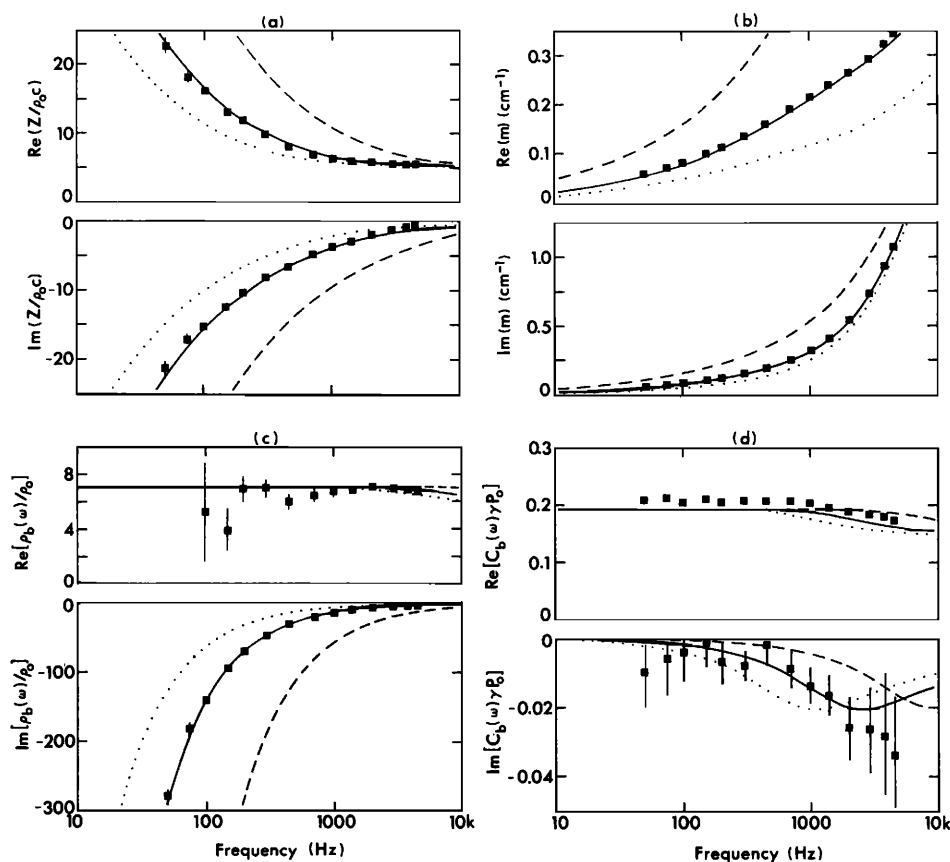


FIG. 5. Comparison of measured (square symbols) and theoretical (solid curves) acoustical properties for a porous material containing 12 111 uniform rectangular pores, each 0.0146×0.0172 cm, giving a porosity of 0.194. The four panels show the same functions as explained in the caption of Fig. 4. The dashed curve is the theoretical prediction that would be obtained if the pores had dimensions of 0.0073×0.0086 cm, and the dotted curve is the prediction for pore dimensions of 0.0219×0.0258 cm.

ance Z and propagation constant m are shown in panels (a) and (b); the complex density and compressibility functions for the bulk material, $\rho_b(\omega)$ and $C_b(\omega)$, are shown in panels (c) and (d), respectively. For all four functions, the real and imaginary components are displayed separately. The triangular symbols show the measured results, with error bars indicating the experimental uncertainty (unless smaller than the symbol size used). Also shown on this figure are the theoretical predictions that follow from Eqs. (30) and (31) for sound propagation in triangular tubes. The solid curve is for the assumed triangle side dimension of 0.0374 cm. For this choice, a porosity of 0.232 and a flow resistivity of 44.7 cgs rayl/cm follow. Very good agreement between theory and experiment is obtained. To illustrate the sensitivity to pore size, similar calculations for side dimensions of 0.017 and 0.057 cm are shown (both with porosity 0.232) as the dashed and dotted curves, respectively; this represents a range considerably larger than would be bracketed by our experimental uncertainties for the side dimension.

In Fig. 5, the measured and calculated acoustical quantities are shown for the model material that contains rectangular pores. Experimental results have been indicated by the square symbols. The solid curves show the theoretical results assuming pore dimensions of 0.0146×0.0172 cm, with a porosity of 0.194 and a flow resistivity of 107.5 cgs rayl/cm. Good agreement between theory and experiment is found. The dashed and dotted curves correspond to calculations using pore dimensions of 0.0073×0.0086 cm and

0.0219×0.0258 cm, respectively, both using a porosity of 0.194.

The acoustical properties of these model porous materials are completely specified by *either* the characteristic impedance and propagation constant *or* the complex density and compressibility functions. Some general observations about the behavior in the low- and high-frequency regimes can be made. At low frequencies, the characteristic impedance (both real and imaginary components) varies as $\omega^{-1/2}$; at high frequencies, it approaches q^2/Ω . The real and imaginary components of the propagation constant are equal in magnitude and vary as $\omega^{1/2}$ at low frequencies; at high frequencies, the real component (the attenuation) continues to vary as $\omega^{1/2}$, but the imaginary component approaches ω/c . The bulk complex density is dominated by its imaginary component at low frequencies, varying as ω^{-1} ; the real component is constant in both frequency limits (approaching $\rho_0 q^2/\Omega$ at high frequencies). The real part of the bulk compressibility function is significantly larger than the imaginary part at all frequencies. It varies from Ω/P_0 at low frequencies to $\Omega/(\gamma P_0)$ at high frequencies.

IV. USE OF SHAPE FACTORS

In most practical cases, the detailed pore shape of a sample is either not known or too complicated to treat in appropriate mathematical detail. Hence, approximate generalized theoretical descriptions of acoustical propagation are intro-

duced. Additional nonacoustical parameters, such as dc flow resistivity, porosity, and tortuosity, can be measured separately and utilized by the models to predict the acoustical performance of a material. In certain theoretical models, a *shape factor* is introduced to allow for the deviation of the pore shape from a baseline shape, e.g., a circular cross section. Two schemes for the introduction of shape factors will be examined. The effectiveness of these approaches will be evaluated by comparing their predictions to the exact solutions available for materials having pores of uniform cross section.

For these comparisons, it will be of use to have general expressions for the complex density and compressibility functions in various frequency regimes, and these will be developed first.

If a dc pressure difference ΔP is maintained over a length Δz of pore, the average flow velocity within the pore is given by

$$\Delta P = -\sigma_x \Delta z \langle v \rangle, \quad (40)$$

where σ_x is the dc flow resistivity within the pore. It will prove useful to introduce a dimensionless permeability coefficient k_0 : In line with Carman¹⁹ we set

$$\sigma_x = 4k_0 \eta / r_h^2, \quad (41)$$

where the hydraulic radius r_h of the pore is

$$r_h = 2A / \Gamma, \quad (42)$$

and A and Γ are the cross-sectional area and circumference of the pore. The hydraulic radius is a measure of the size of the pore. In terms of the dimensions indicated in Fig. 2, the hydraulic radius is r for circular pores, $2ab/(a+b)$ for rectangular pores, $2b$ for slits, and $d/\sqrt{12}$ for triangular pores. The parameter k_0 is constant for a given pore shape: Carman lists a value of 2 for circular pores, 3 for slits, 5/3 for triangular (equilateral) pores, and 1.78 for square pores.

The flow resistivity σ of a bulk material containing such pores is, from Eq. (33),

$$\sigma = (q^2/\Omega) \sigma_x. \quad (43)$$

Comparing the definition for flow resistivity to that for the complex density function in Eq. (3), we see that

$$\sigma_x = \lim_{\omega \rightarrow 0} i\omega \rho_x(\omega). \quad (44)$$

At low frequencies, then, $\rho_x(\omega)$ can be written as an expansion in powers of ω :

$$\rho_x(\omega) = \sigma_x / i\omega + k_1 \rho_0 + \dots \quad (45)$$

The constant k_1 is independent of frequency (and real), and depends only on the pore cross-sectional shape. By expanding the exact expressions presented in Sec. I B, it may be shown that k_1 takes values of 4/3, 6/5, and 10/7 for the circular, slit, and triangular (equilateral) cases, respectively. Interestingly, it is noted that

$$k_1 = 2k_0 / (2k_0 - 1), \quad (46)$$

for all three of these geometries, but it is not known whether this result is generally true for all pore cross-sectional shapes. Making use of the general relationship for uniform

tubes, Eq. (8), the low-frequency expansion for the complex compressibility function is

$$C_x(\omega) = (1/\gamma P_0) \{ \gamma - (\gamma - 1) [(i\rho_0 \omega N_{pr} / \sigma_x)] + \dots \}. \quad (47)$$

For high frequencies, we can follow the approach described by Johnson *et al.*,²⁰ with some simplification due to the constant cross sections of the assumed pores. The effect of viscosity is confined to a boundary layer near the surface of the pore. In the interior of the pore, away from this boundary layer, the velocity is essentially a constant v_0 given by

$$i\omega \rho_0 v_0 = - \frac{dp}{dz}. \quad (48)$$

Since the boundary layer thickness approaches zero as frequency is increased, the surface locally can be taken as flat. The velocity near the pore surface is

$$v(x, y) = v_0 [1 - \exp(-K\beta)], \quad (49)$$

where β is the distance from the pore wall to the point (x, y) and

$$K = (i\omega \rho_0 / \eta)^{1/2}. \quad (50)$$

Integrating over the cross section, according to Eq. (2) we find

$$\langle v \rangle = v_0 [1 - \Gamma / (KA)]. \quad (51)$$

Then, Eqs. (3), (48), and (51) may be combined to give the high field expression for the complex density function

$$\rho_x(\omega) \approx \rho_0 [1 + (2/r_h)(\eta/i\omega \rho_0)^{1/2}]. \quad (52)$$

The corresponding complex compressibility function is

$$C_x(\omega) \approx \frac{1}{\gamma P_0} \left[1 + (\gamma - 1) \frac{2}{r_h} \left(\frac{\eta}{i\omega N_{pr} \rho_0} \right)^{1/2} \right] \quad (53)$$

in the high-frequency limit.

A. Attenborough approach

Attenborough³⁻⁵ has developed a model for describing the propagation of sound in rigid-frame, air-filled porous materials. In this model, it is assumed that the complex density and compressibility functions can be approximated by the corresponding functions for circular pores if an appropriate scaling of the arguments of these functions is made. For a single pore, we use

$$\rho_x(\omega) = \rho_0 [1 - 2T(\sqrt{-i\lambda_x}) / \sqrt{-i\lambda_x}]^{-1} \quad (54)$$

and

$$C_x(\omega) = \frac{1}{\gamma P_0} \left(1 + (\gamma - 1) \frac{2T(\sqrt{-iN_{pr}\lambda_x})}{\sqrt{-iN_{pr}\lambda_x}} \right). \quad (55)$$

The corresponding expressions for the bulk medium can be computed through Eqs. (33) and (35), and the characteristic impedance and propagation constant follow using Eqs. (36) and (37). The parameter λ_x is calculated making use of subsidiary, nonacoustical information:

$$\lambda_x = (1/2s_p)(8\rho_0 \omega q^2 / \Omega \sigma)^{1/2}. \quad (56)$$

The tortuosity q^2 , porosity Ω , and flow resistivity σ can be measured independently. Only the "pore shape factor ratio"

s_p cannot be measured, and it must be determined by fitting to measured acoustical data or from preknowledge of the pore structure. The use of this shape factor will be examined critically by making use of the known solutions for acoustical propagation in samples containing uniform pores of specific cross-sectional shape.

Since we are considering an application of this theory to materials having pores of uniform cross section, we can use Eq. (43) to give

$$\lambda_x = (1/2s_p)^2 (8\rho_0\omega/\sigma_x)^{1/2}. \quad (57)$$

The low-frequency expansions of Eqs. (54) and (55) are

$$\rho_x(\omega) = (2s_p)^2 (\sigma_x/i\omega) + \frac{4}{3}\rho_0 + \cdots, \quad (58)$$

$$C_x(\omega) = (1/\gamma P_0) [\gamma - (\gamma - 1)(i\rho_0\omega/4s_p^2\sigma_x) + \cdots]. \quad (59)$$

Comparing the leading term of the expansion for $\rho_x(\omega)$ to the corresponding term in the general expression Eq. (45), it is evident that s_p must equal 0.5 if the low-frequency behavior is to be correctly described. The second term of the expansion will be correct only for pores of circular cross section. The leading term for $C_x(\omega)$ is always correctly given (i.e., $1/P_0$). The second term, comparing to Eq. (47), is correct only if $s_p = 0.5$.

At high frequencies, the function $T[(-i)^{1/2}\lambda_x]$ approaches $(-i)$. We thus obtain the high-frequency behavior

$$\rho_x(\omega) \approx \rho_0 [1 + 2s_p (\sigma_x/2i\omega\rho_0)^{1/2}], \quad (60)$$

$$C_x(\omega) \approx (1/\gamma P_0) [1 + 2s_p (\gamma - 1) (\sigma_x/2i\omega N_{pr}\rho_0)^{1/2}]. \quad (61)$$

By comparing these limiting forms to the general expressions for uniform tubes, Eqs. (52) and (53), the appropriate choice of shape factor ratio may be determined. We find that s_p must satisfy

$$s_p^2 = 2\eta/(\sigma_x r_h^2). \quad (62)$$

Making use of Eq. (41), the required pore shape factor ratio is found to be related to the permeability coefficient k_0 as

$$s_p = (2k_0)^{-1/2}, \quad (63)$$

in the high-frequency limit. Hence, s_p must be 0.5 for circular pore materials, $1/\sqrt{6} = 0.408$ for slit-pore materials, and $\sqrt{0.3} = 0.548$ for triangular pore materials.

In the Attenborough model, then, the pore shape factor ratio s_p must be *frequency dependent*, in general, if both low- and high-frequency limits are to be respected. This point is demonstrated graphically in Fig. 6 by examining the predictions of Attenborough's theory for certain special (hypothetical) materials for which exact predictions can be made. Suppose first that we have a porous material that contains uniform slitlike pores of half-width 0.01 cm, and suppose further that the density and sinuosity of the pores is such that the overall porosity is 0.9 and the tortuosity is 1.44. The flow resistivity of the individual pores is 5.44 cgs rayl/cm and the flow resistivity of the bulk material is 8.71 cgs rayl/cm. The characteristic impedance and propagation constant may be calculated exactly from the theoretical formulation of Sec. I, using Eqs. (16)–(18) and (33)–(37). These calculated functions are shown as the solid curves in panels (a) and (b) of Fig. 6. The same data can be used by the Attenborough model, leading to the predictions shown in Fig. 6(a) and (b)

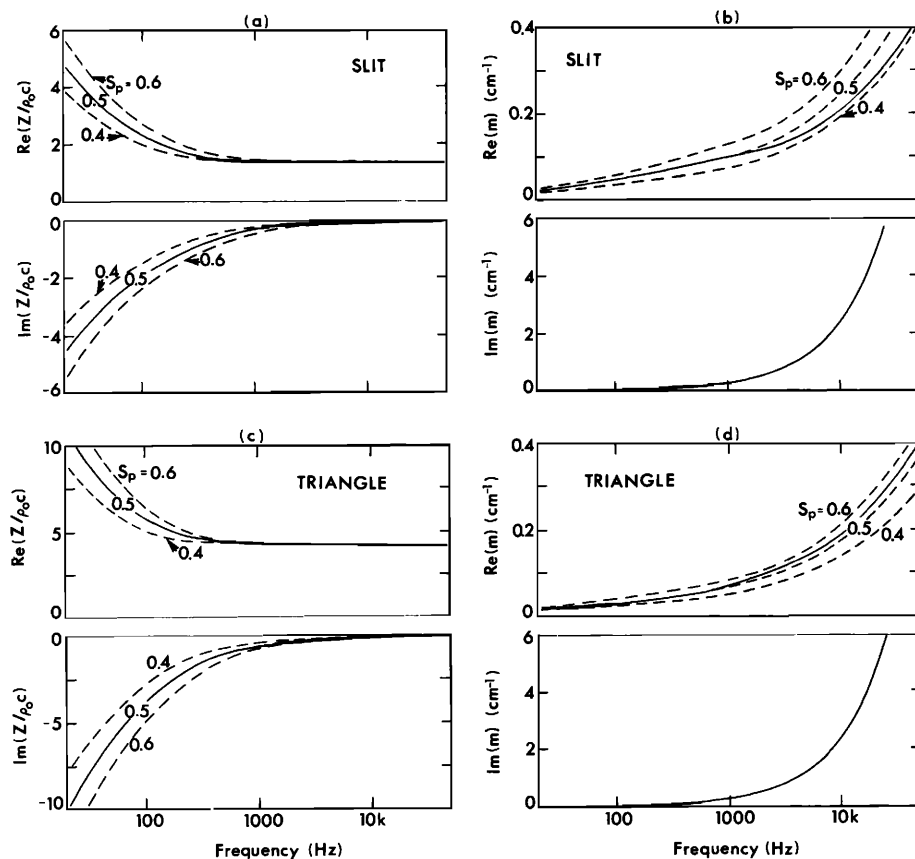


FIG. 6. Comparison of the exact theoretical expressions (solid curves) to the approximate model proposed by Attenborough³⁻⁵ (dashed curves), for porous materials having specific pore shapes. Panels (a) and (b) show the characteristic impedance and propagation constant, respectively, for a material containing slitlike pores of 0.01-cm half-width and having a porosity of 0.9, a tortuosity of 1.44, and a dc flow resistivity of 8.71 cgs rayl/cm. The Attenborough predictions are shown for three choices of "pore shape factor ratio" s_p . Panels (c) and (d) show similar functions for a material containing triangular pores, each with a side of 0.07 cm, and having a porosity of 0.3, a tortuosity of 1.44, and a flow resistivity of 14.2 cgs rayl/cm. For both slitlike and triangular pores, the shape factor ratio must be 0.5 at low frequencies. At high frequencies, the appropriate choice for s_p is 0.408 for slit-type pores and 0.548 for triangular pores.

by the dashed curves. Three choices of s_p , namely 0.4, 0.5, and 0.6, have been used. At low frequencies, a choice of $s_p = 0.5$ is necessary if the characteristic impedance is to be correctly predicted. At high frequencies, this choice would lead to an erroneous prediction of the propagation constant and the choice of 0.408, as discussed above, must be used.

A similar comparison between exact theory and the Attenborough model is shown in panels (c) and (d) of Fig. 6, for a material containing triangular pores of side 0.07 cm. A porosity of 0.3 and tortuosity of 1.44 are used, giving a flow resistivity of 14.2 cgs rayl/cm for the bulk material (2.96 cgs rayl/cm in individual pores). Again, a pore shape factor ratio of 0.5 is required at low frequencies but is inappropriate at high frequencies. A value of $s_p = 0.548$ is required in the high-frequency limit.

B. Generalized Biot approach

We consider here an alternate approach to the introduction of a shape factor, one that does not require that the shape factor be frequency dependent. This is achieved by building the low-frequency limit explicitly into the assumed general form. As suggested by Allard *et al.*,²¹ we set

$$\rho_x(\omega) = \rho_0 + (\sigma_x/i\omega)F(\lambda_x), \quad (64)$$

where the function F is the viscosity correction factor introduced by Biot⁸

$$F(\lambda_x) = -\frac{1}{4} \frac{\sqrt{-i\lambda_x} T(\sqrt{-i\lambda_x})}{1 - 2T(\sqrt{-i\lambda_x})/(\sqrt{-i\lambda_x})}. \quad (65)$$

The scaling parameter λ_x is now defined as

$$\lambda_x = s(8\rho_0\omega q^2/\Omega\sigma)^{1/2}, \quad (66)$$

with s being the new shape factor. If we restrict our attention solely to materials containing uniform pores, then, λ_x is also given by

$$\lambda_x = s(8\rho_0\omega/\sigma_x)^{1/2}. \quad (67)$$

The compressibility function must be defined in a compatible manner. From the general relation in Eq. (8),

$$C_x(\omega) = \frac{1}{\gamma P_0} \left(\gamma - (\gamma - 1) \times \frac{\rho_0}{\rho_0 + \sigma_x F(\sqrt{N_{pr}} \lambda_x)/(i\omega N_{pr})} \right). \quad (68)$$

At low frequencies, we obtain the following expansions for the complex density and compressibility functions:

$$\rho_x(\omega) = \sigma_x/i\omega + (1 + s^2/3)\rho_0 + \cdots, \quad (69)$$

$$C_x(\omega) = (1/\gamma P_0) [\gamma - (\gamma - 1)(i\rho_0 N_{pr} \omega/\sigma_x) + \cdots]. \quad (70)$$

Comparing the expansion for $\rho_x(\omega)$ to Eq. (45), the leading term is seen to be correctly given, *independent* of the choice of shape factor. The difference between the two expansions appears in the second term. Comparing the expansion for $C_x(\omega)$ to Eq. (47), the first two terms are noted as being in agreement. Thus the limiting low-frequency behavior is correctly given by this model, independent of choice of shape

factor. This is a direct result of assuming the basis form of Eq. (64), with σ_x built in.

The choice of shape factor s could be made so that the second expansion term in Eq. (69) is correct. However, it was found to be more important to ensure that the high-frequency behavior is properly described. At high frequencies, we obtain from Eqs. (64) and (68) the limiting forms

$$\rho_x(\omega) \approx \rho_0 [1 + s(\sigma_x/2i\omega\rho_0)^{1/2}] \quad (71)$$

and

$$C_x(\omega) \approx (1/\gamma P_0) [1 + s(\gamma - 1)(\sigma_x/2i\omega N_{pr}\rho_0)^{1/2}]. \quad (72)$$

To correctly match the exact high-frequency expansions of Eqs. (45) and (47), the shape factor must be chosen so that

$$s = (2/k_0)^{1/2}. \quad (73)$$

This result is similar to the earlier result of Eq. (63) for Attenborough's model but as defined, s and $2s_p$ are the inverse of each other [i.e., compare Eqs. (56) and (66)] and the functional forms assumed for $\rho_x(\omega)$ and $C_x(\omega)$ are quite different.

Using the values of k_0 quoted earlier, we would anticipate shape factors of 1.000 for circular pores, 0.816 for narrow slits, 1.095 for triangular (equilateral) pores, and 1.060 for square pores.

The effectiveness of this generalized theoretical approach in representing the acoustical behavior of materials containing uniform pores of arbitrary cross section is investigated in Fig. 7. The known exact solutions for three prototype porous materials (shown as the solid curves) are compared to the predictions of the generalized theory (shown as the dashed curves, except where graphically indistinguishable from the solid curves). Two of the porous materials, one containing slitlike pores and the other containing triangular pores, were described earlier and used to evaluate Attenborough's approach (see Fig. 6). Shape factors of 0.816 and 1.095, respectively, are used in applying the generalized theory. As a third example, we consider a material containing square pores, each 0.025 cm on a side, with a porosity of 0.4, a tortuosity of 1.69, and a flow resistivity of 8.7 cgs rayl/cm. A shape factor of 1.06 is used for this material. For the characteristic impedance and propagation constant [panels (a) and (b)], the generalized Biot approach gives very good agreement with the exact solutions for these three materials. Only in the attenuation function (i.e., the real part of m), for λ_x near unity, can differences in the curves be seen. In panels (c) and (d), the complex density and compressibility functions are seen to give this same good agreement. Some differences are apparent in the real part of the complex density at low frequencies but, because the imaginary component becomes much larger than the real component, these discrepancies will be acoustically insignificant.

The calculations shown in Fig. 7 are representative. Supplementary calculations, using different choices of pore size, porosities, tortuosities, and frequency ranges, give comparably good agreement between our new theoretical approach and the exact solutions. For a given pore shape, the acoustical properties are described quite well using a *single, frequency-independent* shape factor.

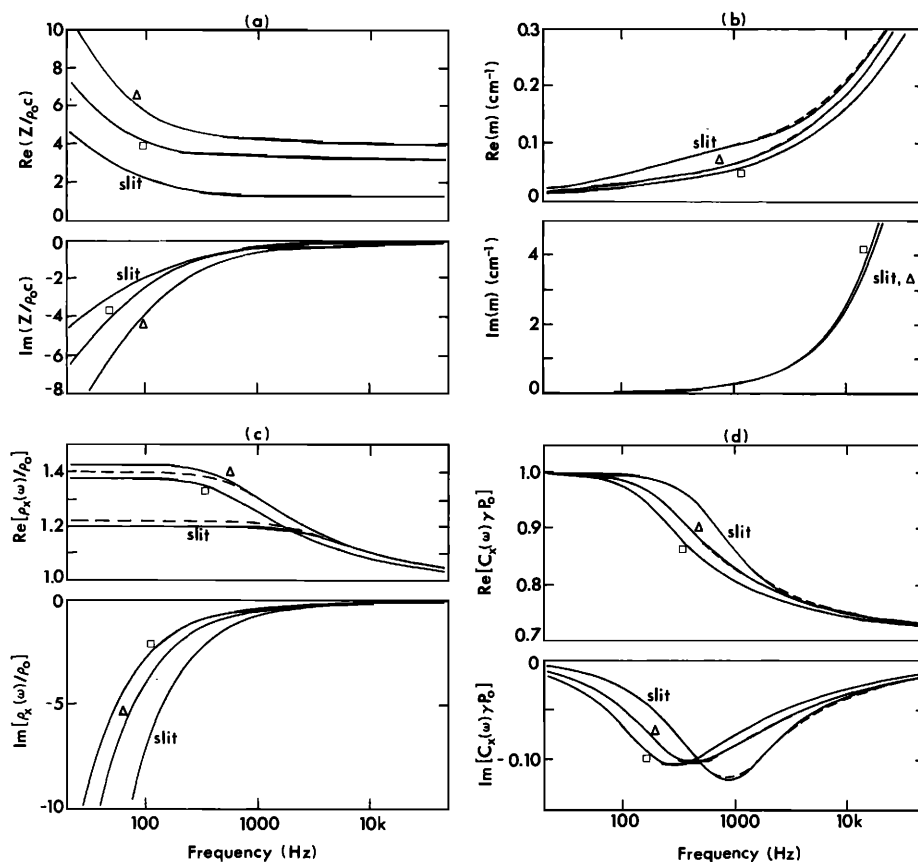


FIG. 7. Comparison of the exact theoretical expressions (the solid curves) to a new approximate model appropriate for uniform tubes (the dashed curves). Panels (a) and (b) show the characteristic impedance and propagation constant, respectively, of the bulk material, while panels (c) and (d) show the complex density $\rho_x(\omega)$ and compressibility $C_x(\omega)$ functions, respectively, within the pores. Three different hypothetical samples are included in these comparisons. Two of these, a sample containing slit-like pores and one containing triangular pores, have been described in the caption of Fig. 6. The third is a sample containing square pores, each 0.025 cm on a side, with a porosity of 0.4, a tortuosity of 1.69, and a flow resistivity of 8.7 cgs rayl/cm. For most of the curves, the approximate model matches the exact curves very well.

V. CONCLUDING REMARKS

Exact solutions have been presented to describe the acoustical behavior of porous materials containing uniform pores of specific cross-sectional shapes. The pore shapes discussed were circular, rectangular (including the limiting case of a narrow slit), and triangular cross sections. Careful measurements on two model porous materials (triangular pores and rectangular pores) have validated the theoretical approach for obtaining these solutions. We note that there is a general relation [i.e., Eq. (8)] between the complex density function and the complex compressibility function, for materials containing pores whose cross-sectional shape does not change along their lengths.

By considering hypothetical porous materials for which exact acoustical solutions exist, two generalized theoretical models have been evaluated. The Attenborough theory³⁻⁵ was found to describe the acoustical properties of these prototype materials only if the "pore shape factor ratio" s_p was allowed to vary with frequency. It was demonstrated that this shape factor must take a value of 0.5 for *all* pore shapes at low frequencies, and a value that depends on the pore shape [according to Eq. (63)] at high frequencies. An alternative theoretical generalization has been introduced. This approach makes use of the viscosity correction function introduced by Biot⁸ but has been extended to include thermal conductivity effects that lead to a frequency-dependent compressibility function. A *single* shape factor was found to give

very good agreement between this theoretical generalization and the exact solutions.

There is a limitation on the immediate applicability of our theoretical formulation and our introduction of a shape factor. In its present form, the theory describes only materials containing pores that are *uniform* in cross-sectional shape and size along their length. Materials for which the pore cross sections change along their lengths are excluded, since Eq. (8) would not be applicable. It is possible to extend the theory, though. For example, pores that do not vary in cross section too rapidly can be approximated by adjoining lengths of uniform tubes of different cross sections. Possible extensions to the theory and parallel experimental effort²² are now under study.

ACKNOWLEDGMENTS

The authors are grateful to Dr. J.-F. Allard, Dr. K. Attenborough, and Dr. G. Daigle for their comments on this work. Contributions by G. Reiser and J. Quaroni to the development of the measurement systems and by R. Lemoine to the fabrication of the model porous materials are appreciated.

¹ M. E. Delany and E. N. Bazley, "Acoustical properties of fibrous materials," *Appl. Acoust.* 3, 105-116 (1970).

² P. M. Morse and K. U. Ingard, *Theoretical Acoustics* (McGraw-Hill, New York, 1968).

- ³ K. Attenborough, "Acoustical characteristics of rigid fibrous absorbents and granular materials," *J. Acoust. Soc. Am.* **73**, 785–799 (1983).
- ⁴ K. Attenborough, "On the acoustic slow wave in air-filled granular media," *J. Acoust. Soc. Am.* **81**, 93–102 (1987).
- ⁵ K. Attenborough and O. Buser, "On the application of rigid-porous models to impedance data for snow," *J. Sound Vib.* **124**, 315–327 (1988).
- ⁶ C. Zwikker and C. W. Kosten, *Sound Absorbing Materials* (Elsevier, Amsterdam, 1949).
- ⁷ M. A. Biot, "Theory of propagation of elastic waves in a fluid-saturated porous solid. I. Low-frequency range," *J. Acoust. Soc. Am.* **28**, 168–178 (1956).
- ⁸ M. A. Biot, "Theory of propagation of elastic waves in a fluid-saturated porous solid. II. Higher frequency range," *J. Acoust. Soc. Am.* **28**, 179–191 (1956).
- ⁹ M. R. Stinson, "The propagation of plane sound waves in narrow and wide circular tubes, and generalization to uniform tubes of arbitrary cross-sectional shape," *J. Acoust. Soc. Am.* **89**, 550–558 (1991).
- ¹⁰ H.-S. Roh, W. P. Arnott, J. M. Sabatier, and R. Raspet, "Measurement and calculation of acoustic propagation constants in arrays of small air-filled rectangular tubes," *J. Acoust. Soc. Am.* **89**, 2617–2624 (1991).
- ¹¹ M. R. Stinson and Y. Champoux, "Assignment of shape factors for porous materials having simple pore geometries," *J. Acoust. Soc. Am. Suppl.* **1 88**, S121 (1990).
- ¹² H. Tidjeman, "On the propagation of sound waves in cylindrical tubes," *J. Sound Vib.* **39**, 1–33 (1975).
- ¹³ I. S. Gradshteyn and I. W. Ryzhik, *Table of Integrals, Series, and Products* (Academic, New York, 1965), Secs. 1.441 and 1.445.
- ¹⁴ Y. Champoux and M. R. Stinson, "Measurement of the characteristic impedance and propagation constant of materials having high flow resistivity," *J. Acoust. Soc. Am.* **90**, 2182–2191 (1991).
- ¹⁵ J. D. McIntosh, M. T. Zuroski, and R. F. Lambert, "Standing wave apparatus for measuring fundamental properties of acoustic materials in air," *J. Acoust. Soc. Am.* **88**, 1929–1938 (1990).
- ¹⁶ Y. Gibiat and F. Laloë, "Acoustical impedance measurements by the two-microphone-three-calibration (TMTC) method," *J. Acoust. Soc. Am.* **88**, 2533–2545 (1990).
- ¹⁷ M. R. Stinson and G. A. Daigle, "Electronic system for the measurement of flow resistance," *J. Acoust. Soc. Am.* **83**, 2422–2428 (1988).
- ¹⁸ Y. Champoux, M. R. Stinson, and G. A. Daigle, "Air-based system for the measurement of porosity," *J. Acoust. Soc. Am.* **89**, 910–916 (1991).
- ¹⁹ P. C. Carman, *Flow of Gases through Porous Media* (Academic, London, 1956).
- ²⁰ D. L. Johnson, J. Koplik, and R. Dashen, "Theory of dynamic permeability and tortuosity in fluid-saturated porous media," *J. Fluid Mech.* **176**, 379–402 (1987).
- ²¹ J. F. Allard, C. Depollier, J. Nicolas, W. Lauriks, and A. Cops, "Propriétés acoustiques des matériaux poreux saturés d'air et théorie de Biot," *J. Acoustique* **3**, 29–38 (1990).
- ²² Y. Champoux and M. R. Stinson, "Experimental investigation of models of sound wave propagation in air-saturated porous media," *J. Acoust. Soc. Am. Suppl.* **1 88**, S121 (1990).